DICTO 1:2 Truncated Octahedron

The 4D Vacuus of the Permutohedron



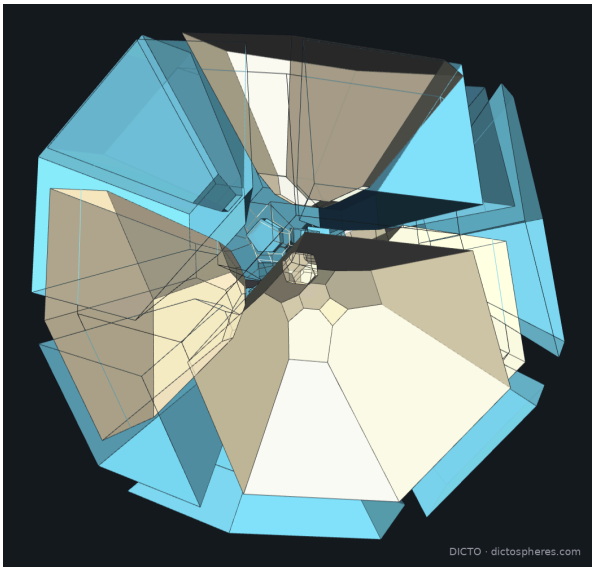
the web of all orderings of five things, its rules made solid: 65 central slices in four shapes, five permutohedra of 1, 2, 4, 5 at the heart slotted along regular hexagons, a count for every dimension, and DICTONIC's golden bond as a walk through permutohedra

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DICTO: ideas, names, constructions and direction.
Computation, verification and drafting with Claude (Anthropic).

Version 3 adds a reference list (each entry checked against its publisher's record), the path of DICTO's work that led here (Section 6), #93's theorem that the DICTO 1:2 and 2:1 fill space together in exactly one way, the resonance arrangement for the two-valued slices, and credits from a gallery search for the figures. No result changes.

Version 2 adds the hero shots from the page *The 4D Permutohedron* (Build, cells pulled apart), Figure 2 (the orderings on every corner, the exchanges on every edge), the cutaways of the heart (Figure 3) and of the DICTO 1:2 Truncated Octahedron's own 3D Vacuus (Figure 4: three octagons in the coordinate planes, twelve hexagons), and names the cells as what they are, permutohedra. No result changes.

Version 1. Theorems 1–4 are proved here, each count also checked by exact computer arithmetic (rational and $\mathbb{Q}(\sqrt{5})$); checked before release by an in-house panel of four automated referees and one independent attack (reports and judgements kept with the paper's data).



The 4D permutohedron seen from just outside one of its 3D permutohedra (a Schlegel view), its 30 cells (10 permutohedra, 20 hexagonal prisms) pulled half apart round the empty centre: from the page *The 4D Permutohedron*. Schlegel views from a truncated-octahedron cell are standard (T. Ruen with R. Webb's Stella4D; H. S. Teoh); we found no published view of the cells pulled apart.

Write the numbers 1 to 5 in every possible order: 12345, 12354, …, 54321. There are 120 orders. Link two of them whenever one becomes the other by swapping two neighbouring values, 2 and 3 say. The result is a web of 120 knots and 240 exchanges, and it is the skeleton of a four-dimensional solid, the 4D permutohedron. Its smallest rings are its faces: squares, where two swaps do not disturb each other, and hexagons, where three swaps go round and come back (Figure 1). This paper builds the 4D Vacuus of that solid: the few flat slices through its centre that between them carry every face, and so every rule of the web.

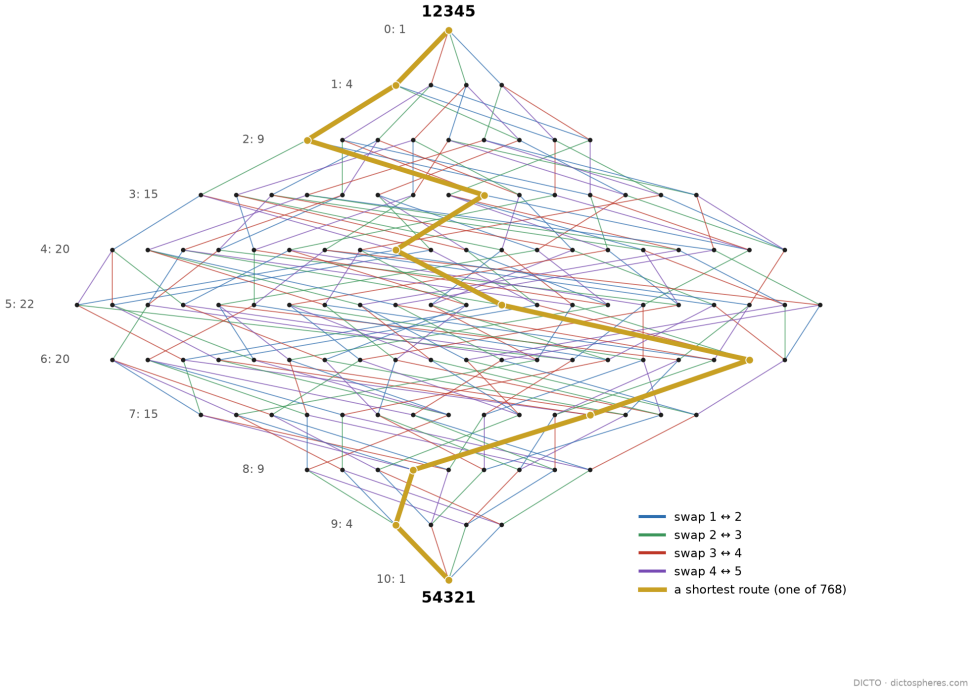


Figure 1. The web of exchange. Rows: the number of pairs out of order (1, 4, 9, 15, 20, 22, 20, 15, 9, 4, 1 orderings). Bond colours: which neighbouring values are swapped. Gold: one of the 768 shortest routes from 12345 to 54321.

1. The web of exchange

The 4D permutohedron is the convex hull of the 120 points (p_1, …, p_5) whose coordinates are 1, …, 5 in some order; it lies in the 4-space $p_1 + \ldots + p_5 = 15$, centred at (3, 3, 3, 3, 3). It is simple (four edges at each corner) and has 240 edges, 150 two-dimensional faces (60 hexagons, 90 squares) and 30 cells: 10 three-dimensional permutohedra of four consecutive values (one coordinate fixed at 1 or at 5; classically the truncated octahedron) and 20 hexagonal prisms. Its edges join orders that differ by swapping two neighbouring values, so its skeleton is the web above: four exchanges at every knot, in ten directions. The following facts are classical and were recounted exactly (web_of_bonds.py): the number of exchanges between two orders is the number of pairs they put in opposite order; 54321 is farthest from 12345, ten exchanges away, along 768 shortest routes (Stanley's count of reduced words of the longest permutation); the 2-faces are the web's rings, a square for two swaps of disjoint value pairs and a hexagon for the braid ring $s_k s_{k+1} s_k = s_{k+1} s_k s_{k+1}$. In #104 the same solid was swept by a moving 3-space: 26 cells are born, 66 pairs swap and 26 cells die, and these are the numbers of orders with one, two and three descents (1, 26, 66, 26, 1): the h-vector of the permutohedron is the Eulerian numbers (Postnikov, Reiner and Williams 2008).

2. The 4D Vacuus

The *Vacuus* of a centrally symmetric polytope is the set of hyperplanes through its centre spanned by its ridges (the faces one dimension below the cells), each ridge lying in exactly one; a *slice* is the polytope cut by one of them. DICTO's Vacuus solids (#87) are built from such central slices (there as fins, one per edge, which for these solids fill central discs): the octahedron from three squares through its centre that carry all its edges, the cuboctahedron from four hexagons. One dimension up the slices are 3D and carry the 2-faces. A 2-face of the 4D permutohedron does not pass through the centre, so together with the centre it spans

exactly one 3-space through the centre: its *central slice*. The 4D Vacuus is the set of these slices.

A 2-face is an ordered split of the positions {1, …, 5} into three blocks B_1, B_2, B_3 of sizes (a, b, c): the positions in B_1 hold the a smallest values in any order, those in B_2 the next b, those in B_3 the largest c. Its central slice has the normal vector that takes the value $c_1 = bc(b + c)$ on B_1, $c_2 = -ac(a + 2b + c)$ on B_2 and $c_3 = ab(a + b)$ on B_3 (it is orthogonal to (1, …, 1), constant on each block, and orthogonal to the vector of centred block sums $S_1 = -a(b + c)/2$, $S_2 = b(a - c)/2$, $S_3 = c(a + b)/2$, so its hyperplane passes through the face; checked against exact elimination for every 2-face).

Theorem 1 (the 4D Vacuus). The 150 two-dimensional faces of the 4D permutohedron lie in exactly 65 central slices, each face in one. Up to the symmetries of the solid (the 120 permutations of the coordinates and the central inversion) there are four slices:

slices	split (a, b, c)	normal (values on B_1 | B_2 | B_3)	slice shape	edge lengths²	faces of the 4D solid carried
30	(1, 2, 2)	(8 | −7 −7 | 3 3)	32 corners, 52 edges; 14 four-sided faces (2 of them squares) + 8 hexagons	eight values, 13/18 to 8	2 squares
20	(1, 1, 3)	(6 | −9 | 1 1 1)	truncated-octahedron type: 6 rectangles + 8 hexagons (2 regular)	2 and 56/9	2 hexagons
10	(1, 3, 1)	(3 | −2 −2 −2 | 3)	24 corners, 42 edges; 18 four-sided faces (6 squares, 12 trapezoids) + 2 regular hexagons	2, 4, 8	2 hexagons
5	(2, 1, 2)	(1 1 | −4 | 1 1)	the DICTO 1:2 Truncated Octahedron (permutohedron of 1, 2, 4, 5): 6 squares + 8 hexagons	2 and 8	6 squares

Proof. The normal formula shows that a slice depends only on the blocks and the values (c_1, c_2, c_3); since $c_1 > 0 > c_2$, for five values two splits give the same slice only if they are reverses of each other (B_3, B_2, B_1), or, when a = c, have the same middle block (for an even number of values there is one more coincidence, Theorem 3). Counting: splits (1, 1, 3) and (1, 2, 2) give 5!/(1!1!3!) = 20 and 5!/(1!2!2!) = 30 slices; (1, 3, 1) and (2, 1, 2) give C(5, 3) = 10 and C(5, 1) = 5; in all 65, and 30·2 + 5·6 = 90 squares, 20·2 + 10·2 = 60 hexagons. Slices with the same split are images of one another under a permutation of the coordinates, so they are congruent. The shapes were computed in exact rational arithmetic: corners are the solid's corners on the slice and the points where edges cross it; every face proposed by a hull routine was confirmed exactly (all its corners on one plane, every other corner strictly on one side), and every shape has Euler number 2. ∎

An earlier floating-point count split the 30 congruent slices of the first row into several apparent types; the exact computation above supersedes it.

3. The heart: five permutohedra of 1, 2, 4, 5

The five slices of the last row are the slices $p_i = 3$: the middle value held in one place while the other four take 1, 2, 4, 5 in every order. Each is the 3D permutohedron of the values 1, 2, 4, 5: the **DICTO 1:2 Truncated Octahedron**, a truncated octahedron in shape (the same faces meeting in the same way as Archimedes', but not his equal-edged solid) with squares of side 1 and hexagons whose sides are 1 and 2 in turn (√2 and 2√2 in these coordinates), and it carries the six squares of the 4D solid in which the middle value stays put. These five are the heart of the Vacuus (Figure 3). Sweeps of sections parallel to a cell, which pass through such

a slice, are known (J. Bowers; Galoomba, Polytope Wiki); we found none that names the central one as the permutohedron of 1, 2, 4, 5 or shows the five together.

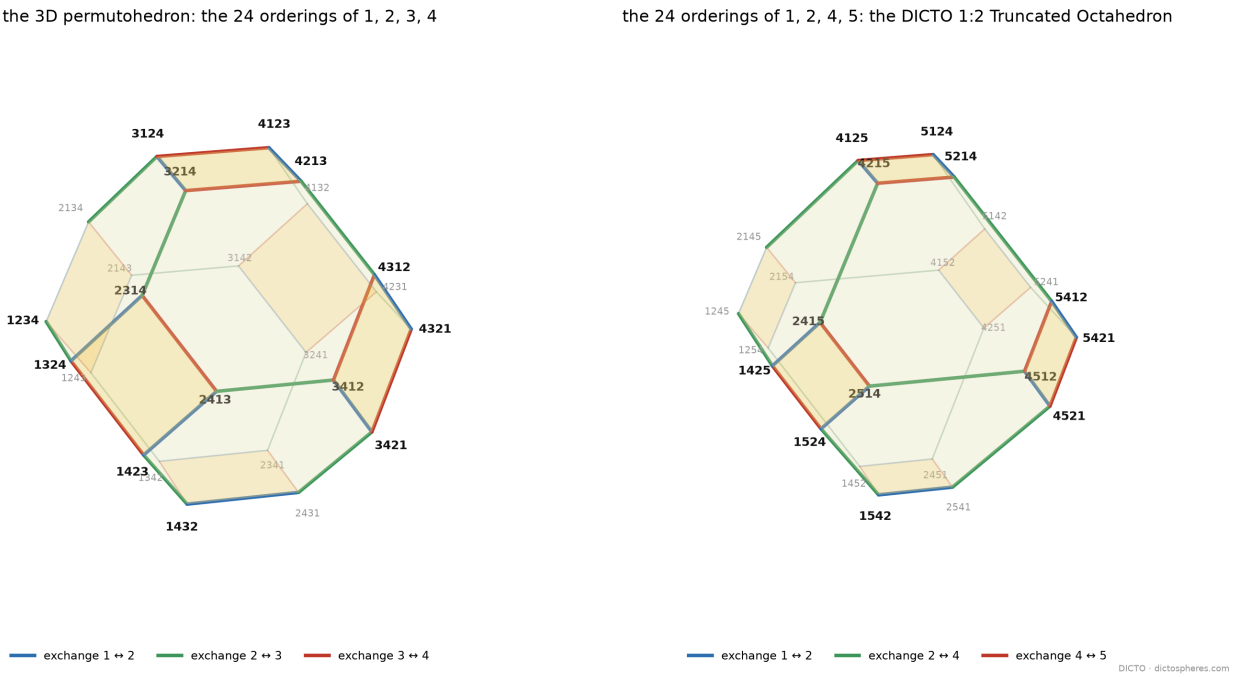
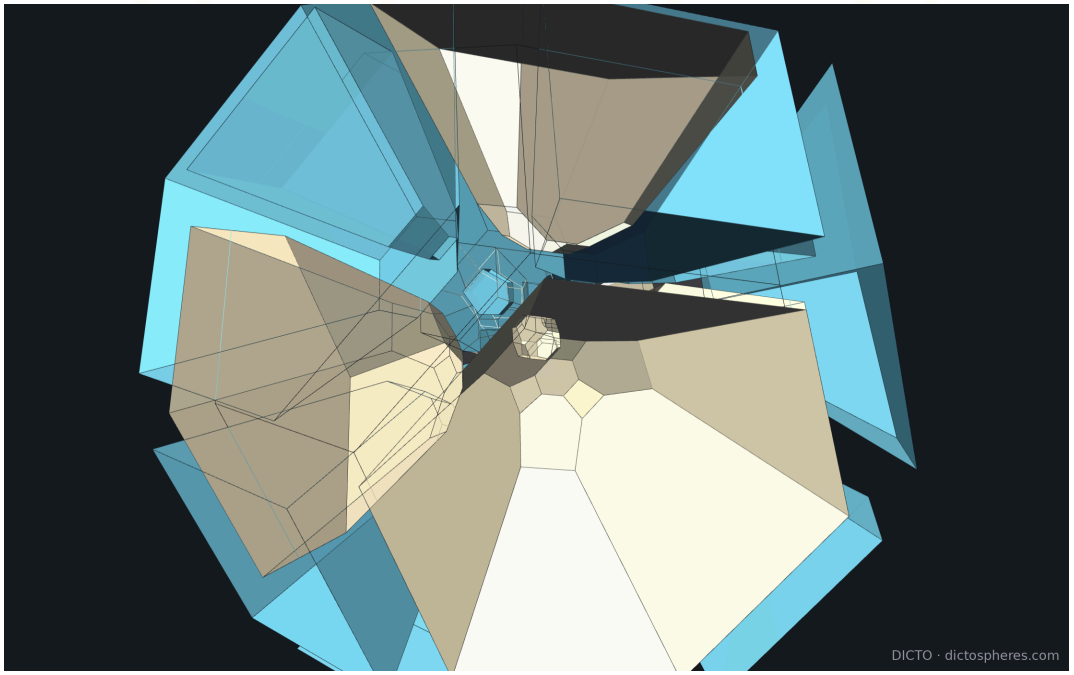


Figure 2. The permutohedron is its orderings. Left: the 3D permutohedron, the 24 orderings of 1, 2, 3, 4, each edge an exchange of two neighbouring values. Right: the 24 orderings of 1, 2, 4, 5, the same form with the exchange 2 ↔ 4 drawn long: the DICTO 1:2 Truncated Octahedron, five of which make the heart of the 4D Vacuus. Near corners in bold. (Permutohedra labelled by orderings, edges by exchanges, are a standard picture: T. Piesk, Wikimedia Commons, 2020; D. Eppstein.)



Close up: into the hollow centre of the pulled-apart 4D permutohedron, where the smallest cells of the view sit.

Theorem 2 (how the heart is slotted together). Any two of the five central permutohedra, $p_i = 3$ and $p_j = 3$, cross in a regular hexagon through the centre, with side equal to its radius $2\sqrt{2}$; its corners are the points with $p_i = p_j = 3$ and the other three coordinates 1, 3, 5 in any order. Three of them share a segment through the centre, and four or five share only the centre.

Proof. A point lies in the permutohedron exactly when its coordinates sum to 15 and its k largest coordinates sum to at most 5 + 4 + … (the k largest values), for every k (majorization; Rado 1952, after Hardy, Littlewood and Pólya). With $p_i = p_j = 3$ and the other three q_1, q_2, q_3 summing to 9, these conditions reduce to $1 \le q_m \le 5$: a hexagon with corners the orders of (1, 3, 5), each at distance $\sqrt{8}$ from the centre and from its neighbours. The plane meets no corner or edge of the solid; its six corners are where it pierces 2-faces, which was confirmed exactly by intersecting the plane with all 150 faces. Three conditions $p_i = p_j = p_k = 3$ leave $q_1 + q_2 = 6$ with $1 \le q \le 5$, a segment; four leave the centre. ∎

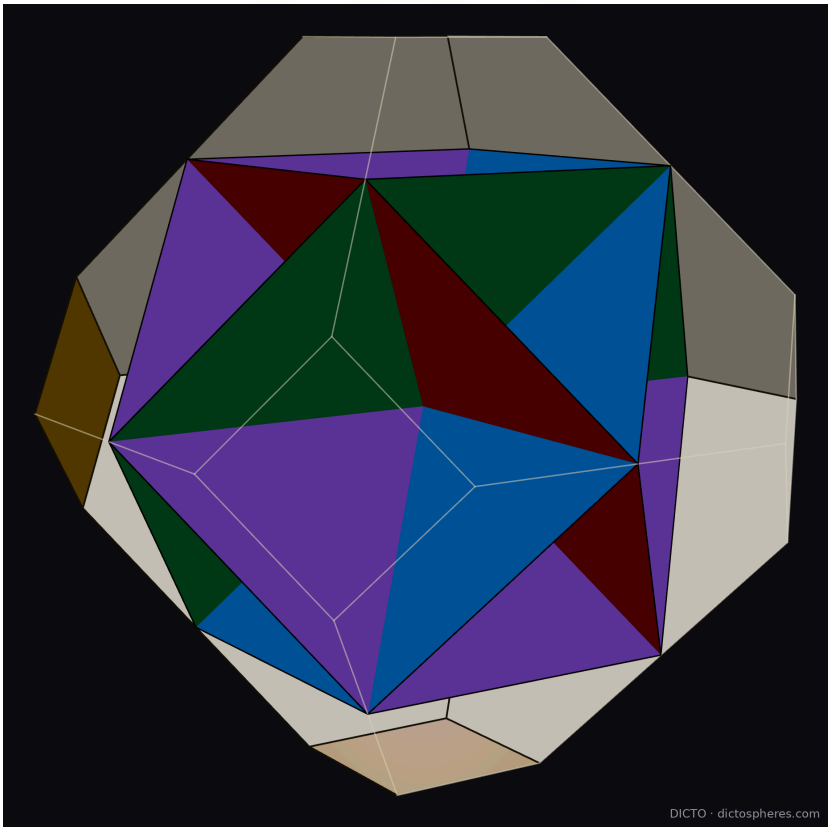


Figure 3. One of the five central slices, the permutohedron of 1, 2, 4, 5, cut away to its back half (its squares, gold, are faces of the 4D solid); the four coloured regular hexagons through its centre are where the other four central slices cross it. The cardboard octahedron Vacuus slots three squares along their diagonals; the 4D heart slots five permutohedra of 1, 2, 4, 5 along ten regular hexagons.

The heart nests the dimensions inside one another: the 4D permutohedron of 1, …, 5 holds at its centre, five times, the 3D permutohedron of the four values that remain when the middle one is taken out; in the same way the hexagon of the orders of 1, 2, 3 holds the segment between the orders of 1 and 3. This works whenever the number of values is odd, so that there is a middle value to hold still. (For any value v the hyperplane $p_i = v$ meets no edge inside it, since an edge changes two coordinates from k, k + 1 to k + 1, k through non-

integers; so the slice $p_i = v$ is the permutohedron of the other $n - 1$ values, and it passes through the centre exactly when v is the middle value.)

4. The Vacuus in every dimension

The construction works for the permutohedron of 1, …, n in dimension $n - 1$: its ridges (faces one dimension below the cells) are the ordered splits into three blocks, and each spans one central slice with the centre.

Theorem 3 (the Vacuus count). The ridges of the permutohedron of 1, …, n lie in exactly N(n) central hyperplanes, where
$N(n) = \Sigma_{a<c}\, n!/(a!\, b!\, c!) + T(n)$, the sum over a, b, c ≥ 1 with $a + b + c = n$, and $T(n) = 2^{n-1} - 1$ for n odd, $2^{n-2} - 1$ for n even.

n (dimension)	3 (2)	4 (3)	5 (4)	6 (5)	7 (6)	8 (7)	9 (8)
ridges	6	36	150	540	1806	5796	18150
Vacuus slices N(n)	3	15	65	225	770	2443	7761

Proof. By the normal formula, a split with $a \neq c$ gives a slice whose normal takes three different values; the normal is unique up to scale, its level sets are the blocks, its one negative level is B_2, and the two positive levels are B_1 and B_3 in either order, so only the split and its reverse give the slice, and no two-valued slice can equal it: the first sum. A split with $a = c$ gives a normal with two values, c_1 on $S = B_1 \cup B_3$ (of even size 2a) and c_2 on the middle block T; the ratio $c_1 : c_2 = -b : 2a$ depends only on the sizes of S and T, so the slice is fixed by the unordered pair {S, T}, and every pair with |S| even and at least 2 occurs. When n is odd exactly one of S, T has even size: $2^{n-1} - 1$ pairs. When n is even, $|S| = 2a$ is even so |T| is even too, and the split (a, b, a) gives the same slice as (b/2, 2a, b/2) with S and T exchanged: the pairs with both parts even, $(2^{n-1} - 2)/2 = 2^{n-2} - 1$. The two-valued slices are the hyperplanes $\Sigma_{i \in S}\, x_i = 0$ through the centre (centred coordinates), parallel to facets: members of the resonance (all-subsets) arrangement (Kamiya, Takemura and Terao 2011; Billera et al. 2012; Kühne 2023). The three-valued slices are not of that kind. In closed form, with c_n the central trinomial coefficient (1, 1, 3, 7, 19, 51, …; OEIS A002426), the splits with $a = c$ number $A_n = c_n - 1 - [n \text{ even}] \cdot C(n, n/2)$, and $N(n) = (3^n - 3 \cdot 2^n + 3 - A_n)/2 + T(n)$. Brute force over every ridge for n = 3, …, 9, with each slice found by exact elimination, gives the same numbers (vacuus_general.py). ∎

In two dimensions the Vacuus of the hexagon is its three long diagonals; in three, the 3D permutohedron's 36 edges lie on 15 planes through its centre. So do the 36 edges of the DICTO 1:2 Truncated Octahedron: its 15 cards are three octagons in the three coordinate planes, slotted at right angles like the three squares of the octahedron Vacuus, each carrying 4 edges, and twelve hexagons carrying 2 each (Figure 4; vacuus3d_12.py, exact). Neither N(n) nor its three-valued part (0, 12, 50, 210, 707, 2380, 7506) is in the On-Line Encyclopedia of Integer Sequences (searched 11 November 2026 for 3, 15, 65, 225, 770, 2443 and for 12, 50, 210, 707, 2380); we make no stronger claim of novelty.

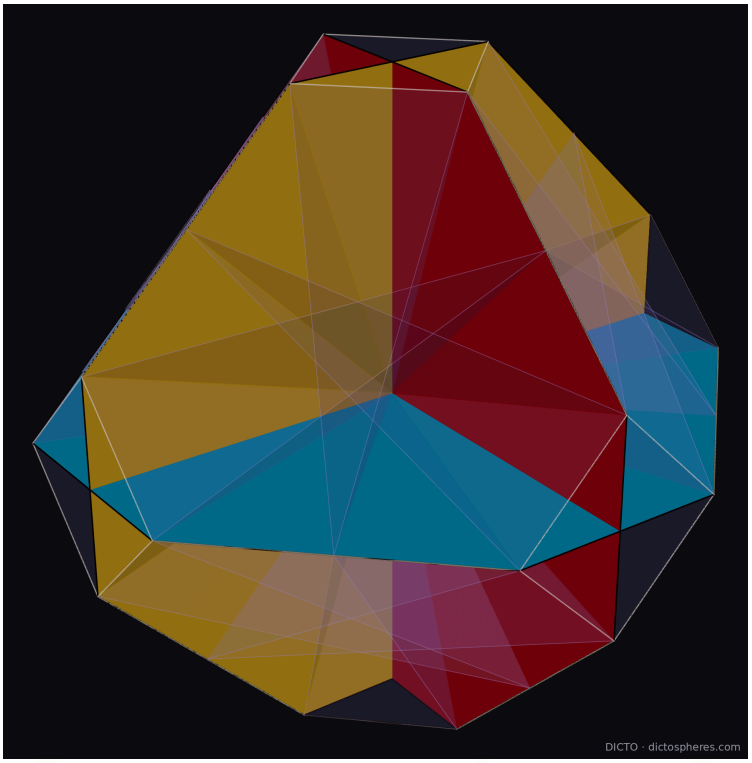


Figure 4. The 3D Vacuus of the DICTO 1:2 Truncated Octahedron: its 36 edges on 15 cards through the centre. The three octagons (gold, red, blue) lie in the coordinate planes and slot together at right angles; the twelve hexagon cards are drawn as faint glass.

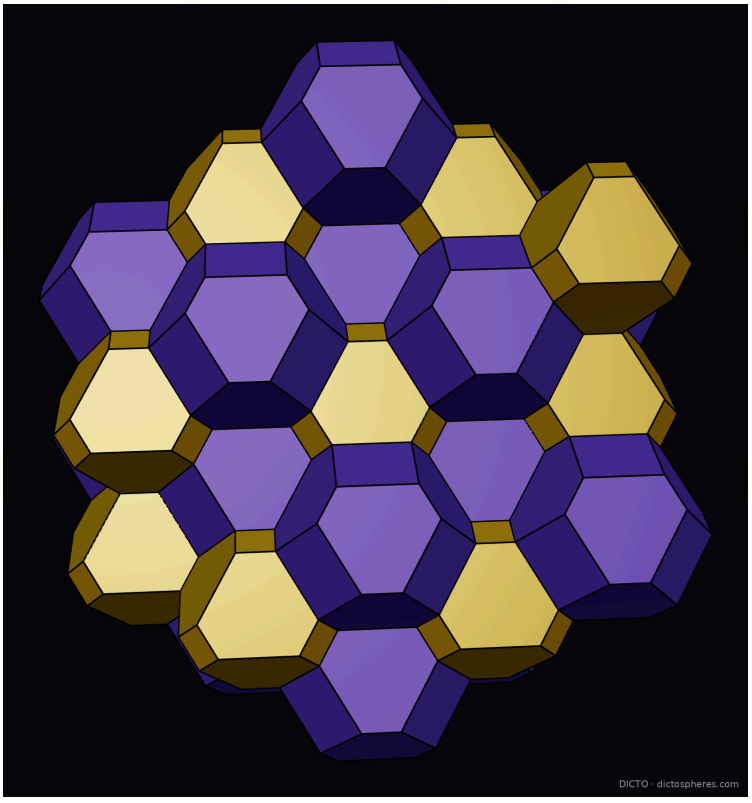
5. DICTONIC: the golden bond walks through permutohedra

DICTONIC's golden bond (#48, Theorem 4) uses the family of semi-uniform truncated octahedra T(x), the hull of all signed orders of (1 + x, x, 0) (Koca et al. 2015: the orbit of the octahedral group with Dynkin labels (1, x, 0)), which with T(1/x) on the cube centres fills space. It is a walk through permutohedra. At x = 1/2 the partner T(2) is the **DICTO 2:1 Truncated Octahedron**: squares of side 2, hexagons with sides 2 and 1 in turn, the permutohedron of −5, −1, 1, 5; the two solids' hexagons are congruent, and along a shared hexagon each edge of the 1:2 between two hexagons meets an edge of the 2:1 between a hexagon and a square. The 1:2 alone cannot fill space by translations: its hexagons are not centrally symmetric (side 1 is opposite side 2), and a convex solid that fills space by translations has centrally symmetric facets (Minkowski for lattices; Venkov, McMullen in general). Whether turned and mirrored copies can do it is not settled here. With its partner the answer is complete: in #93 the pieces P_μ and $Q_\mu = P_{3-\mu}$ have squares of edge $\sqrt{2}(\mu - 1)$ and hexagon sides alternating with $\sqrt{2}(2 - \mu)$, so at $\mu = 4/3$ they are the DICTO 1:2 and the DICTO 2:1, and #93's Theorem A says that every tiling by the two is the caesium-chloride one: they fill space together in exactly one way.

Three solids. Read by the ratio of square side to hexagon side, the slider holds three simplest members: the 1:1 (the 3D permutohedron itself, Kelvin's truncated octahedron, x = 1), which fills space alone and is its own partner; and the DICTO 1:2 and DICTO 2:1 Truncated Octahedra (x = 1/2 and x = 2), mirror images of each other in proportion, which fill space only together. (A 2:2 is the 1:1 at twice the size: only the ratio decides the shape.) Every other ratio on the slider pairs with its reverse in the same way.

Theorem 4. For every $x > 0$, $T(x)$ is, up to scale, the permutohedron of four values placed symmetrically about zero, $-q, -p, p, q$, with $x = (q - p)/(2p)$. In particular: $x = 1$ (Kelvin's truncated octahedron) is the permutohedron of $-3, -1, 1, 3$, equally spaced, the 3D permutohedron itself; $x = 1/2$ is the permutohedron of $-2, -1, 1, 2$, which is $1, \ldots, 5$ without its middle value: the DICTO 1:2 Truncated Octahedron, the heart of the 4D Vacuus (Theorem 2); and $x = \tau$ (the golden bond) is the permutohedron of $-\tau^3, -1, 1, \tau^3$. Equivalently, $T(x)$ is the orbit of the weight with A_3 Dynkin labels $(x, 1, x)$, whose full octahedral symmetry is the diagram symmetry of equal outer labels.

Proof. The map $y \mapsto \tfrac{1}{2}(y_1 + y_2 - y_3 - y_4, y_1 - y_2 + y_3 - y_4, y_1 - y_2 - y_3 + y_4)$ is an isometry of the 3-space $y_1 + \ldots + y_4 = 0$ onto ordinary space. It sends the order $(q, p, -p, -q)$ to $(p + q, q - p, 0)$, and the 24 orders of $-q, -p, p, q$ to the 24 signed orders of $(p + q, q - p, 0)$, which are the corners of $T(x)$ scaled by $(q - p)/x$ when $(1 + x)/x = (p + q)/(q - p)$, that is $x = (q - p)/(2p)$. For $x = \tau$, $q = p(1 + 2\tau) = p\tau^3$. Checked exactly for $x = 1/2$, 1 and τ by comparing all 276 distances between corners (dictonic_family.py). ∎



DICTO 1:2 Truncated Octahedra (gold) fill space with their partners, DICTO 2:1 Truncated Octahedra (violet; squares of side 2, hexagons with sides 2 and 1 in turn): the stop $x = 1/2$ of DICTONIC's golden bond (Section 5). Unlike the 1:1, the 1:2 does not fill space by shifting alone; its partner is the 2:1.

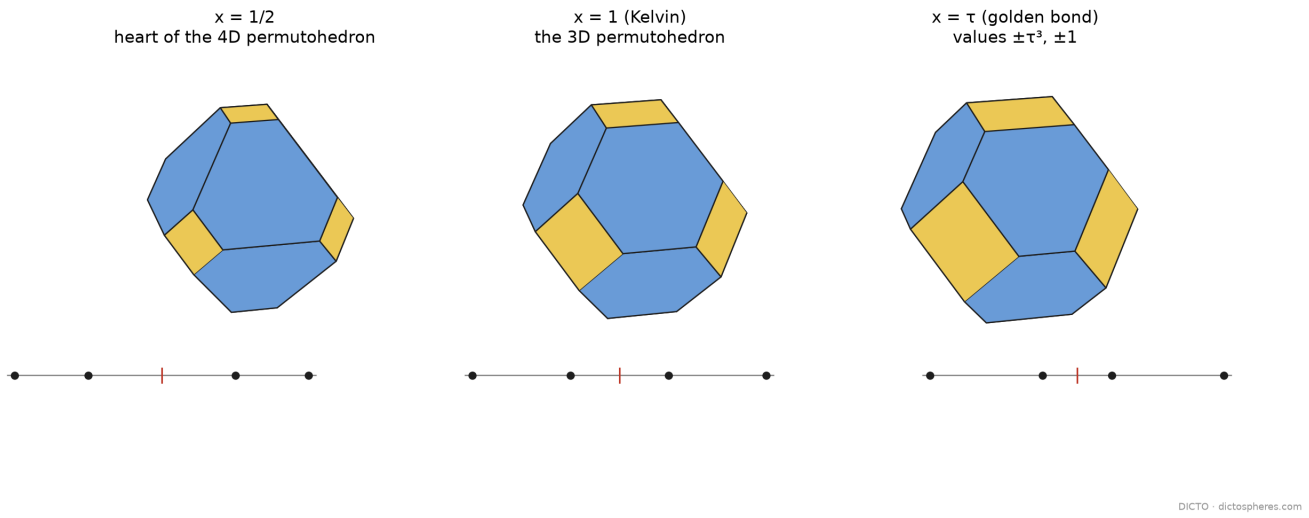


Figure 5. Three stops of DICTONIC's golden slider, each the permutohedron of the four values on the line beneath it (the red mark is the centre). At x = 1/2 the values are 1, ..., 5 without the middle: the slider passes through the heart of the 4D permutohedron.

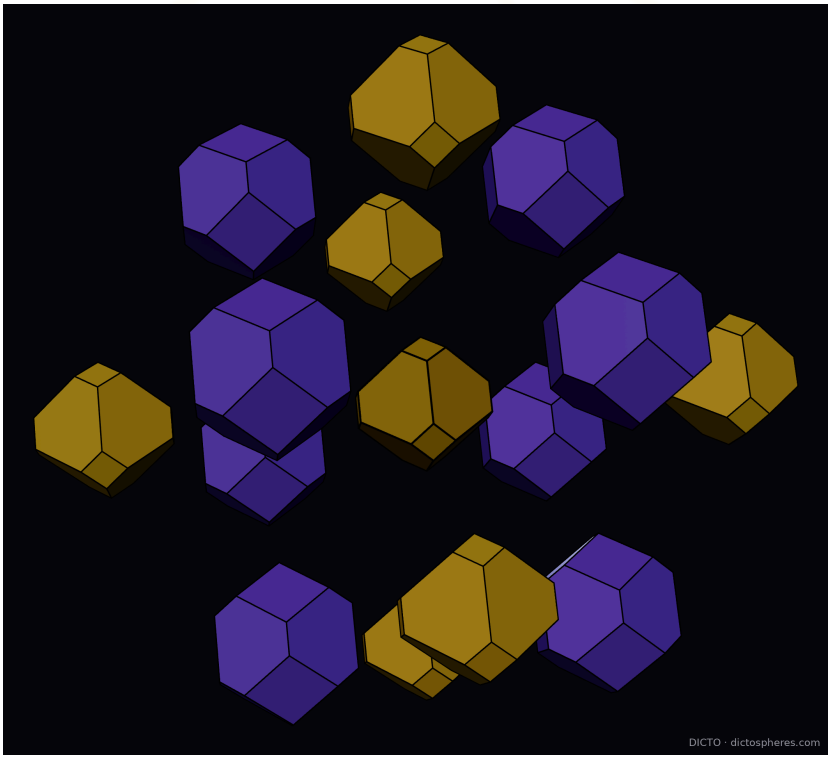


Figure 6. The golden-bond filling at x = 1/2 pulled apart round one DICTO 1:2 Truncated Octahedron (gold, centre): its eight DICTO 2:1 partners (violet) and its six nearest 1:2 neighbours, each moved straight out from the centre.

6. The path here

The DICTO 1:2 Truncated Octahedron grew from two lines of DICTO's work. The first began in Polyhedraverse, where rhombic dodecahedra cut in half and rejoined gave a slightly irregular truncated octahedron that still filled space (#58). #48 then placed truncated octahedra in a family T(x), each filling space with its partner T(1/x), from Kelvin's cell at x = 1 to the golden pair, the Icosaxes and the Unborn Axes, at x = τ; the DICTO 1:2 is its member x = 1/2. #52 read the family as one weight sliding between the two kinds of site of the caesium-chloride network, and #93 proved that each pair on the walk fills space in that one way only.

The hexagons of the DICTO 1:2 are truncated triangles, the face that #88 found running through the golden cuts: there their sides are in the ratio φ, here in the ratio 2; and #88 and #89 found the golden pairs as slices of four-dimensional solids. The second line is the Vacuus: #44 cut the Platonic solids from their centres into Aristotle's pyramids, and #87 kept only the walls between those pyramids, one fin on every edge, so that the octahedron's Vacuus is three squares through its centre and the cuboctahedron's four hexagons. This paper joins the two: one dimension up, the walls become three-dimensional central slices of the 4D permutohedron, and the five at its heart are DICTO 1:2 Truncated Octahedra, the stop x = 1/2 of the slider of #48.

7. Leads

(1) The Vacuus of other simple 4-polytopes, the Queens of #104 first: how many central slices carry their faces, and which of them are 3D solids already in the work. (2) The golden slice of the permutohedron honeycomb (the 4D permutohedra fill 4-space) as a non-repeating 3D filling: its shapes counted exactly. (3) The heart for seven values: the 6D permutohedron holds seven 5D permutohedra of 1, 2, 3, 5, 6, 7; how they are slotted. (4) A physical build of the heart: five card permutohedra cannot be slotted in ordinary space, but their ten crossing hexagons can be shown in the app.

References

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